



NYC Yellow Taxi – Big Data Analytics

BITS ZG522 · Big Data Systems · Assignment 1 · 3 months (Jan–Mar 2024) · ~9M trips

Total Trips

8,479,421

Revenue (USD)

\$235.12 M

Avg Fare

\$33.70

Avg Tip %

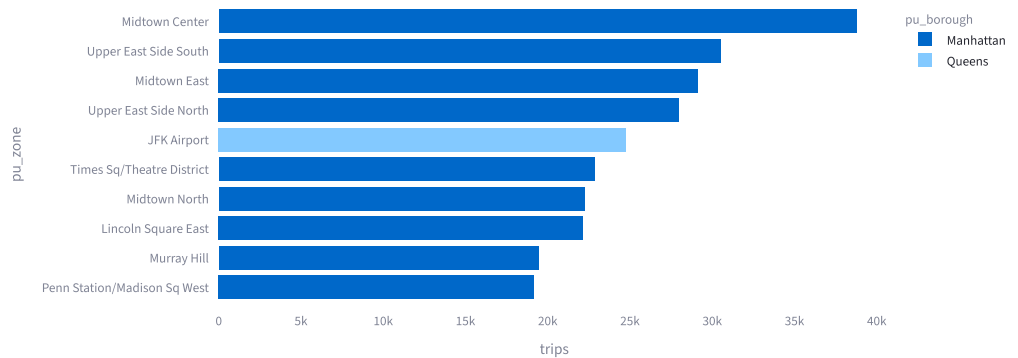
4.7%

Q1 · Demand hotspots by hour

Pickup hour

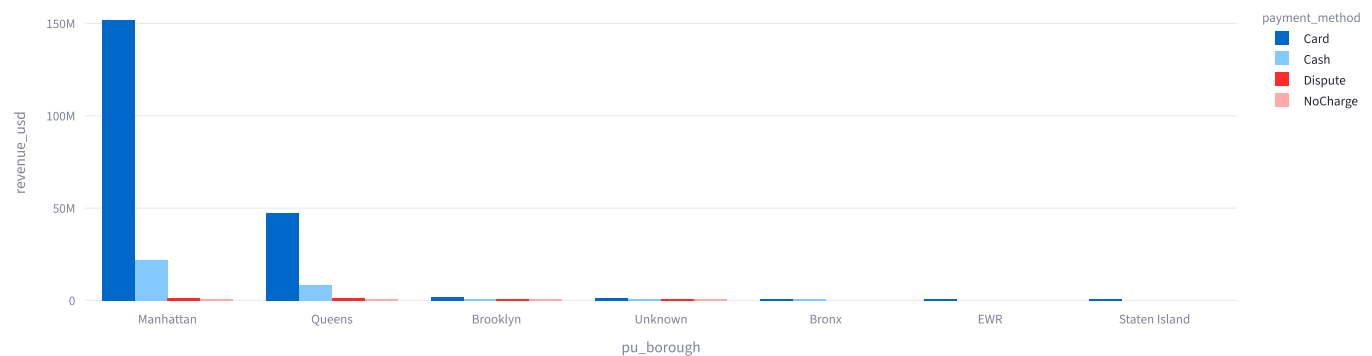


Top 10 pickup zones at 18:00



Q2 · Revenue by borough and payment method

Revenue (USD) by borough × payment



Q3 · Congestion – minutes per mile by hour

Avg minutes per mile across all busy PU→DO pairs



Top 15 slowest PU→DO pairs (min per mile)

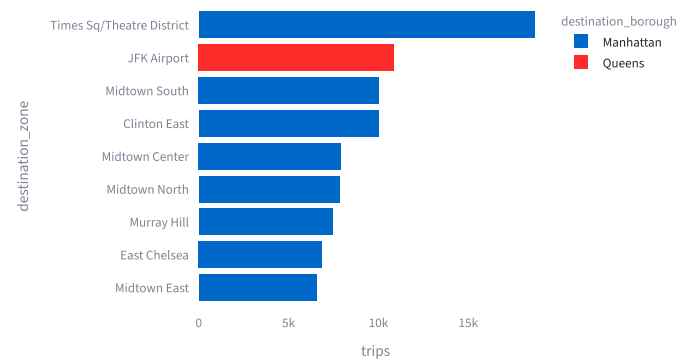
	pickup_hour	pu_loc_id	do_loc_id	trips	avg_duration_min	avg_distance_mi	min_per_mile
432	10	186	230	1147	17.33	1.08	17.16
133	13	186	230	1660	16.12	1.03	16.67
198	11	186	230	1476	16.56	1.07	16.67
123	12	186	230	1693	16.46	1.05	16.49
347	17	161	230	1229	10.74	0.74	15.39
441	16	186	230	1141	14.99	1.05	14.83
362	11	186	161	1217	18.28	1.28	14.73
332	10	186	161	1242	18.07	1.27	14.72
284	14	186	230	1298	15.1	1.06	14.7
382	19	161	230	1188	10.02	0.74	14.43

Q4 · Airport trip profile

Airport

☒ JFK ☐ LGA ☐ EWR

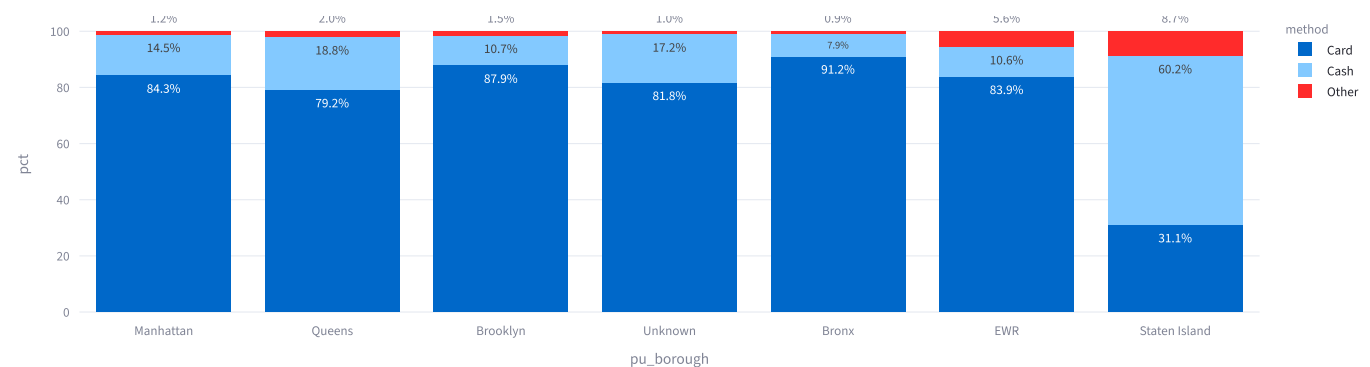
Top destinations from JFK



	airport	destination_zone	destination_borough	trips	avg_distance_mi	avg_
24	JFK	Times Sq/Theatre District	Manhattan	18696	17.99	7
25	JFK	Outside of NYC	None	16343	20.66	10
26	JFK	JFK Airport	Queens	10876	2.79	3
27	JFK	Midtown South	Manhattan	10034	17.28	7
28	JFK	Clinton East	Manhattan	10000	18.25	7
29	JFK	Midtown Center	Manhattan	7916	17.72	7
30	JFK	Midtown North	Manhattan	7836	18.33	7
31	JFK	Murray Hill	Manhattan	7448	16.92	7
32	JFK	East Chelsea	Manhattan	6847	18.11	7
33	JFK	Midtown East	Manhattan	6555	17.32	7

Q5 · Card vs cash share by borough

Payment method share (%)



Pipeline: HDFS → Pig (ETL) → native MR (trips-per-zone) → Hive (analytics) → HBase (zone lookup) → Streamlit

